

Afonso Teodoro – Résumé

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Experience

Jun 2019 - Present Nimble Portal
 Artificial Intelligence Consultant

Engineered and deployed impactful machine learning solutions across computer vision and natural language processing, seamlessly integrating them into mobile and web applications.

- Developing a mobile application for digital car inspections, leveraging TensorFlow and PyTorch models, Docker for deployment, and a React.js annotation platform, alongside an Android proof-of-concept.
- Building Python endpoints for automatic medical expense reimbursement using PyTorch and FastAPI, deployed with Docker.
- Implementing automatic call center transcriptions with Python and Whisper, enabling deeper insights through LLMs.

Part-time endeavours: Developed a market sentiment model by leveraging tweets and Reddit posts and consulted for a Data science platform for medical imaging AI - from labelling images and 3D volumes to integrated model training.

Jun 2015 - May 2019 Instituto de Telecomunicações
 PhD Scholarship

Developed machine learning and optimization methods to tackle a variety of important image processing problems, such as image deblurring, image super-resolution, MRI reconstruction, and image fusion with application to remote sensing.

Dec 2014 - May 2015 Instituto de Telecomunicações
 Research Fellowship for MSc holder

Project PAConvex - *Beyond convexity: non-convex optimization and game-theoretic approaches for imaging inverse problems*

Investigated and tested various methods and tricks to improve the performance of a Gaussian mixture model based image denoising algorithm developed during my MSc thesis.

Feb 2013 - Jun 2013 INESC-ID
 Scientific initiation scholarship

Project ARGUS - *Activity recognition and object tracking based on multiple models*

Developed an interface for trajectory monitoring and detection of abnormal events.

Education

- 2015-2019** PhD Electrical and Computer Engineering - Instituto Superior Técnico
19/20 average score
Thesis: *Mixture Models for Inverse Imaging Problems - Going Beyond Denoising in a Plug-and-Play Fashion*
Supervisor: Professor Mário Figueiredo
- 2012-2014** MSc Electrical and Computer Engineering - Instituto Superior Técnico
18/20 average score
Thesis: *Single-frame Image Denoising and Inpainting using Gaussian Mixture Models*
Supervisors: Professor Mário Figueiredo and Doctor Mariana Almeida

Awards

- IBM Scientific Prize 2019 for "Adapt to survive – Targeted priors in plug-and-play image restoration".

Publications

1. A. Teodoro, M. Almeida and M. Figueiredo. "Single-Frame Image Denoising and Inpainting using Gaussian Mixtures", Int. Conf. Pattern Recognition: Applications and Methods, 2015.
2. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "Image restoration and reconstruction using variable splitting and class-adapted image priors", IEEE Int. Conf. Image Processing, 2016.
3. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "Scene-adapted plug-and-play algorithm with convergence guarantees", IEEE 27th Int. Workshop Machine Learning for Signal Processing, 2017.
4. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "A Convergent Image Fusion Algorithm Using Scene-Adapted Gaussian-Mixture-Based Denoising", IEEE Trans. Image Proc., vol. 28, no. 1, pp. 451–463, Jan 2019.
5. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "Image Restoration using Targeted Plug-and-Play Priors", IEEE Trans. Comp. Imaging, vol. 5, no. 4, pp. 675–686, Dec 2019.
6. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "Block-Gaussian-mixture priors for hyperspectral denoising", Workshop on Signal Proc. with Adaptive Sparse Structured Representations, 2019.
7. A. Teodoro, J. Bioucas-Dias, M. Figueiredo. "Block-Gaussian-Mixture Priors for Hyperspectral Denoising and Inpainting". IEEE Trans. Geosci. and Remote Sens., vol. 59, no. 3, pp. 2478-2486, March 2021.

Skills

■ Programming Languages and Software

Advanced knowledge of MATLAB, Python
Experience with Deep Learning frameworks, PyTorch, Tensorflow, Keras
Experience with LLMs, prompting, Langchain
Knowledge of HTML, Javascript, ReactJS, CSS, Docker
Basic knowledge of Android development, Java, Kotlin
Microsoft Office, L^AT_EX

■ Languages

Native Portuguese
Excellent knowledge of English